

Prerequisite Configuration Steps

Step 1: Install .NET SDK

- **Download and Install .NET SDK:**
- Ensure that the .NET SDK version 8.0.404 is installed on your system. This is crucial for building and running the DataHousekeepingJob application.

Action:

- Download from the official .NET website:
- [Download .NET SDK 8.0.404](#)
- Follow the installation instructions for your OS (Windows in this case).

Verify Installation:

Run the following command in your terminal to confirm the SDK is installed:

```
dotnet --version
```

Step 2: Set the Environment Variable

- **Action:**

Add the environment variable ARCON_CI_CONFIG_PATH_DBCLN to your system. This variable will specify the base path for your configuration files (e.g., nlog.json, internal logs, etc.).

For Windows:

- Open the **Start Menu** and search for **Environment Variables**.
- Click on **Edit the system environment variables**.
- In the **System Properties** window, click **Environment Variables**.
- Under **System variables**, click **New** and add:
 - **Variable Name:** ARCON_CI_CONFIG_PATH_DBCLN
 - **Variable Value:** Path where your configuration files and logs are stored (e.g., C:\MyApp\Config).

Verify the Environment Variable:

Open a new command prompt window and check if the environment variable is set correctly:

```
echo %ARCON_CI_CONFIG_PATH_DBCLN%
```

Step 3: Execute the MSSQL Script

- **Action:**

Navigate to the DataHousekeepingJob\Database_Scripts\MSSQL folder and execute the provided MSSQL script on your database. This script will set up the necessary tables needed for the DataHousekeepingJob to function.

Steps:

- Open SQL Server Management Studio (SSMS) or your preferred SQL client.
- Connect to the target database.
- Open the MSSQL script file located in
DataHousekeepingJob\Database_Scripts\MSSQL.
- Execute the script to create necessary database objects.

Tip: Make sure the user executing the script has the necessary privileges to modify the database schema.

Step 4: Add Common Configuration in common-config.json

- **Action:**

Add the following configuration settings for the DataHousekeepingJob in your common-config.json file.

```
{  
  "datahousekeeping": {  
    "batchsize": 10000,  
    "batchdelay": 500,  
    "scheduledhour": 22,  
  }
```

```

    "scheduledminute": 5
  }
}

```

Explanation:

- **batchsize**: Defines the number of records to process in one batch.
- **batchdelay**: Sets the delay in milliseconds between each batch processing.
- **scheduledhour**: Defines the hour when the housekeeping job should run (in 24-hour format).
- **scheduledminute**: Defines the minute when the job should run.

Ensure this file is saved and correctly located based on your ARCON_CI_CONFIG_PATH_DBCLN environment variable.

Step 5: Configure NLog in nlog-config.json

- **Action:**

Add following NLog configuration to your nlog-config.json file. This will set up logging for the background service.

```

{
  "dbbackgroundservice": {
    "NLog": {
      "internalLogLevel": "Info",
      "internalLogFile":
"${environment:ARCON_CI_CONFIG_PATH_DBCLN:whenEmpty=Log}/Mircoservice_
InternalLog/DataHousekeepingJob/nlogSchedulerService.txt",
      "extensions": [
        { "assembly": "NLog.Extensions.Logging" },
        { "assembly": "NLog.Web.AspNetCore" }
      ],
      "targets": {
        "allfile": {
          "type": "File",
          "fileName":

```

```

"${environment:ARCON_CI_CONFIG_PATH_DBCLN:whenEmpty=Logs}/Microservice
_Logs/DataHousekeepingJob/${shortdate}.txt",
    "layout": "${longdate}|${event-
properties:item=EventId_Id}|${uppercase:${level}}|${logger}|${message}
${exception:format=toString}",
    "maxArchiveDays": 5
  }
},
"rules": [
  {
    "logger": "*",
    "minLevel": "Info",
    "writeTo": "allfile"
  },
  {
    "logger": "Microsoft.*",
    "maxLevel": "Error",
    "final": "true"
  }
]
}
}
}

```

Explanation:

- Internal logs will be saved in the folder of ***Microservice_Logs/DataHousekeepingJob***.
- Logs will be written to files under Microservice_Logs folder with daily rotation.
- The log file format includes timestamp, log level, logger name, message, and exception details.

Save the configuration file in the appropriate location as per your system setup.

Step 6: Install the DataHousekeepingJob as a Service

- **Action:**

Use the `sc.exe` command to install the DataHousekeepingJob as a Windows service. Replace "Path" with the actual path to the executable for the DataHousekeepingJob.

Command:

```
sc.exe create "DataHousekeepingJob" binpath=
"C:\path\to\DataHousekeepingJob\Release\net8.0\DataHousekeepingJob"
start=auto
```

Explanation:

- "DataHousekeepingJob": This is the name of the service.
- `binpath`: Path to the executable that should be run as a service.
- `start=auto`: Configures the service to start automatically when the system boots.

Step 7: Take a Database Backup

- **Action:**

Before starting the DataHousekeepingJob, **ensure a full backup** of your database is taken to avoid any data loss in case of failure during the housekeeping job.

Final Checklist

- .NET SDK 8.0.404 is installed on the system.
- The `ARCON_CI_CONFIG_PATH_DBCLN` environment variable is set.
- The MSSQL script has been executed on the database.
- The `common-config.json` file has been updated with correct data housekeeping configuration.
- The `nlog-config.json` file has been configured for logging.
- The DataHousekeepingJob has been installed as a service using `sc.exe`.
- A database backup has been taken before starting the job.

After completing these steps, you can safely start the DataHousekeepingJob and monitor its performance.